Statistical Analysis of Global Population, Health and Economic Indicators

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1. Introduction

This study examines selected population, health, and economic indicators across countries using the country_data.csv dataset. The analysis focuses on population characteristics, life expectancy, infant mortality, fertility, and GDP per capita. R statistical software will be used to explore the data through descriptive statistics, graphical methods, and appropriate statistical techniques.

Objectives of the Study

The objectives of the study are as follows;

```
I.   To examine the distribution of life expectancy, fertility rates, and infant mortality rates across
II.  To evaluate the association between GDP per capita and life expectancy.
III. To determine the relationship between population growth rates and fertility rates.
IV.  To assess the relationship between infant mortality rates and life expectancy.
V.   To compare differences in life expectancy between male and female populations across countries.
VI.  To examine the relationship between fertility rates and infant mortality rates.
```

2. Data Description

Data Source

The dataset used in this study is country_data.csv, obtained from Kaggle. It contains information on 204 countries and 38 variables covering population, health, economic, education, employment, and environmental indicators. Each observation represents a country, while each variable represents a specific country characteristic. The analysis will focus on selected population, health, demographic, and economic variables relevant to the research questions.

Sample Size

The dataset contains 204 observations representing countries and 38 variables.

Key Variables

Table 1: Description of Variables

Variable	What.it.represents	Role…Type
name	Country	Identifier
region	Geographical region	Categorical
pop_growth	Population growth rate	Numerical
life_expectancy_male	Male life expectancy	Numerical

Variable	What.it.represents	Role…Type
life_expectancy_female	Female life expectancy	Numerical
infant_mortality	Infant mortality rate	Numerical
fertility	Fertility rate	Numerical
gdp_per_capita	GDP per person	Numerical

Data cleaning

The dataset was examined for duplicate observations before conducting the statistical analysis. Twelve duplicate observations were identified and removed from the dataset. This reduced the number of observations from 204 to 192 countries. Removing exact duplicates ensures that individual countries are not unintentionally given greater weight in the subsequent analyses.

Missing Data

```
## # A tibble: 8 x 3
##   variable               missing percentage
##   <chr>                    <int>      <dbl>
## 1 name                         0       0
## 2 region                       0       0
## 3 gdp_per_capita               1       0.52
## 4 life_expectancy_male         5       2.6
## 5 life_expectancy_female       5       2.6
## 6 fertility                    5       2.6
## 7 infant_mortality             7       3.65
## 8 pop_growth                   0       0
```

Missing values were assessed for the variables included in the analysis. The results showed relatively low levels of missing data. GDP per capita had one missing observation (0.52%), while male life expectancy, female life expectancy, and fertility each had five missing observations (2.60%). Infant mortality had seven missing observations (3.65%), while population growth had no missing observations. Because the proportion of missing values was low, observations were retained and analyses were conducted using the available observations for the variables required for each analysis.

3. Methods

The dataset was examined in R to assess its structure, variable types and data quality. Duplicate observations were removed, and the selected variables were checked for missing and unusual values. Missing observations were excluded only from analyses involving the affected variables.

Descriptive statistics were used to summarise male and female life expectancy, fertility rate and infant mortality. Density plots and histograms were used to examine the distributions of these variables across countries.

Spearman's rank correlation was used to examine associations between GDP per capita and life expectancy, population growth and fertility, infant mortality and life expectancy, and fertility and infant mortality. This method was chosen because it measures the strength and direction of monotonic associations without requiring normally distributed variables

Male and female life expectancy were compared using a paired analysis. The Shapiro–Wilk test was used to assess the normality of the differences. As the differences were not normally distributed, the Wilcoxon signed-rank test was used to determine whether there was a statistically significant difference between male and female life expectancy.

4. Results

I. Distributions of health and demographic indicators

What are the distributions of life expectancy, fertility and infant mortality across countries?

Mean life expectancy was 69.6 years for males (median 70.1, IQR 10.4) and 74.5 years for females (median 76.2, IQR 11.5). Female life expectancy was left-skewed, with its mean below its median, indicating a tail of countries with much lower life expectancy than the typical country; male life expectancy was only mildly skewed. Life expectancy ranged from 50.4 to 81.7 years for males and from 54.8 to 87.5 years for females.

Fertility rates ranged from 1.1 to 7.0 births per woman, with a median of 2.3 (IQR 1.85), close to the replacement level of about 2.1. The distribution was right-skewed (mean 2.76), as most countries had low to moderate fertility while a smaller group had very high rates. Infant mortality ranged from 1.3 to 81.9 deaths per 1,000 live births. It was strongly right-skewed: the median was 14.6 but the mean was 22.2, and the standard deviation (19.8) was almost as large as the mean. Half of the countries had infant mortality below 14.6, while a smaller group recorded rates several times higher.

Because several of these distributions are skewed, the median and interquartile range describe the typical country better than the mean and standard deviation. This also supports the use of rank-based methods in the analyses that follow.

Table 2: Summary statistics of the analysed variables

Variable	n	Mean	SD	Median	IQR	Min	Max
Male life expectancy (years)	187	69.6	7.4	70.1	10.4	50.4	81.7
Female life expectancy (years)	187	74.5	7.8	76.2	11.5	54.8	87.5
Fertility (births per woman)	187	2.8	1.3	2.3	1.9	1.1	7.0
Infant mortality (per 1,000)	185	22.2	19.8	14.6	29.3	1.3	81.9
Population growth (%)	192	1.3	1.1	1.1	1.5	-1.5	4.3
GDP per capita	191	15586.4	25775.0	5794.6	15256.7	99.6	185835.0

Figure 1 shows the distribution of life expectancy, Figure 2 fertility rates and Figure 3 infant mortality.

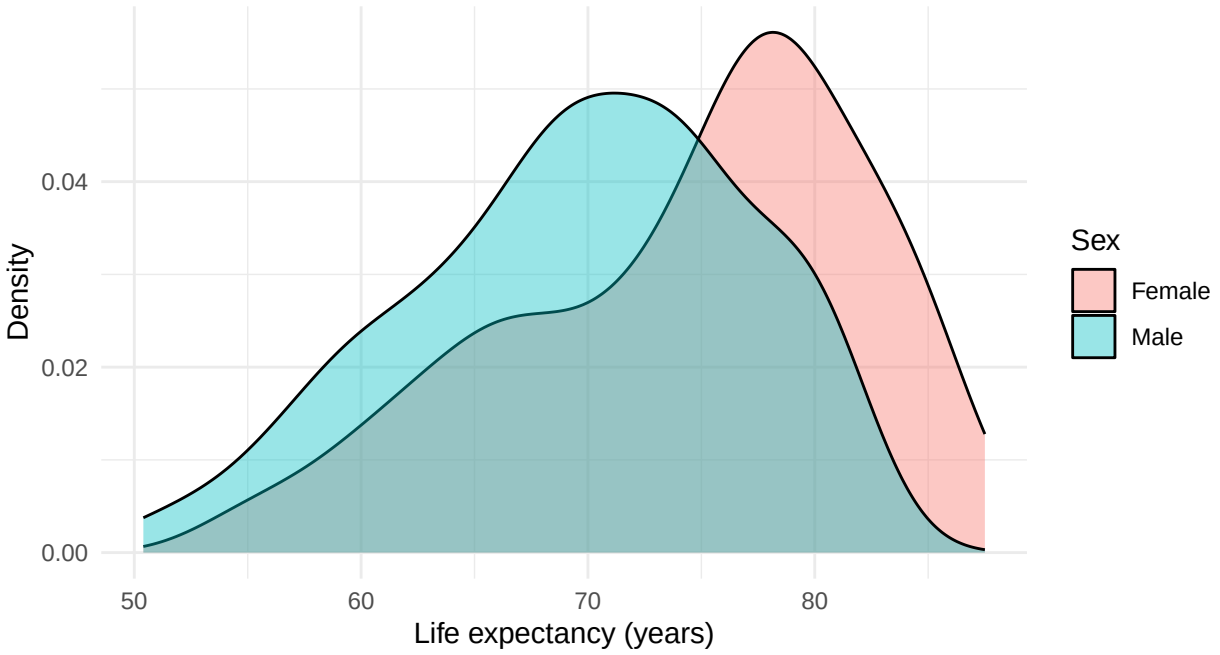


Figure 1: Distribution of male and female life expectancy across countries

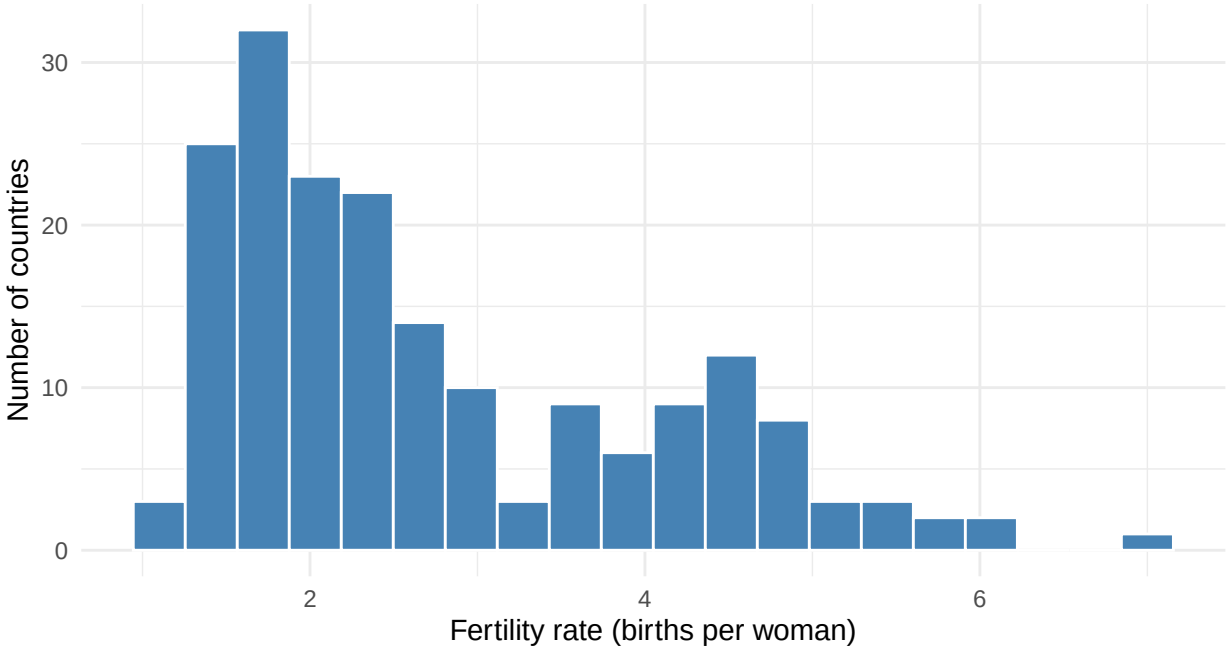


Figure 2: Distribution of fertility rates across countries

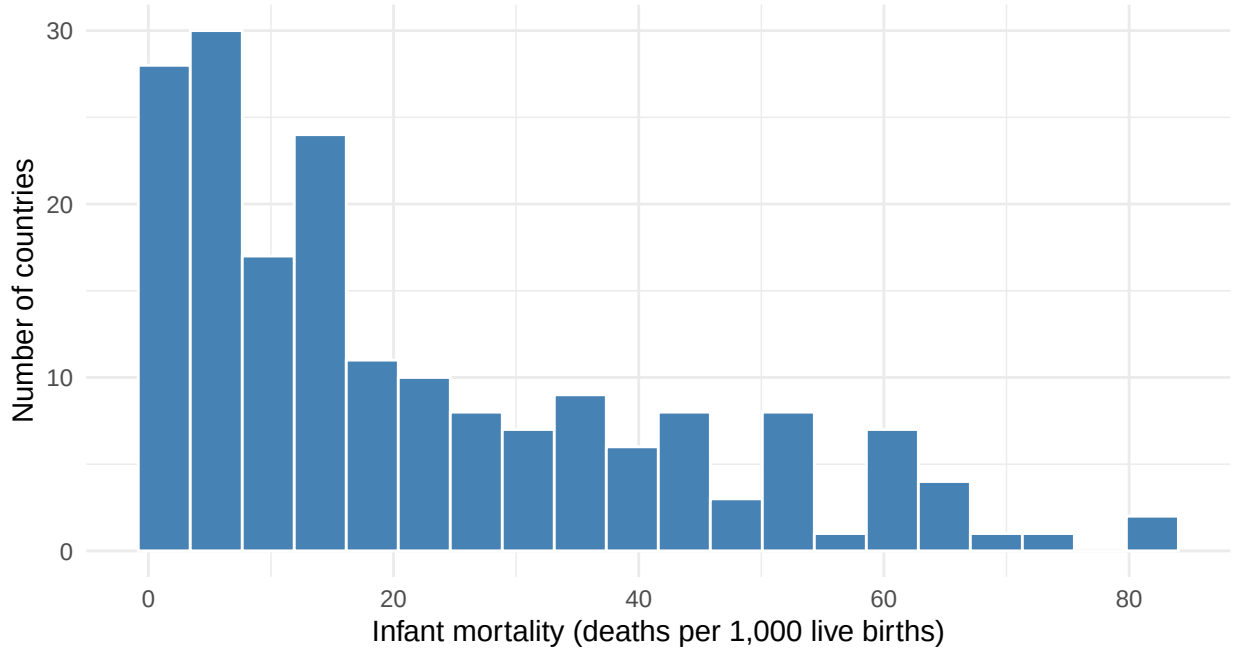


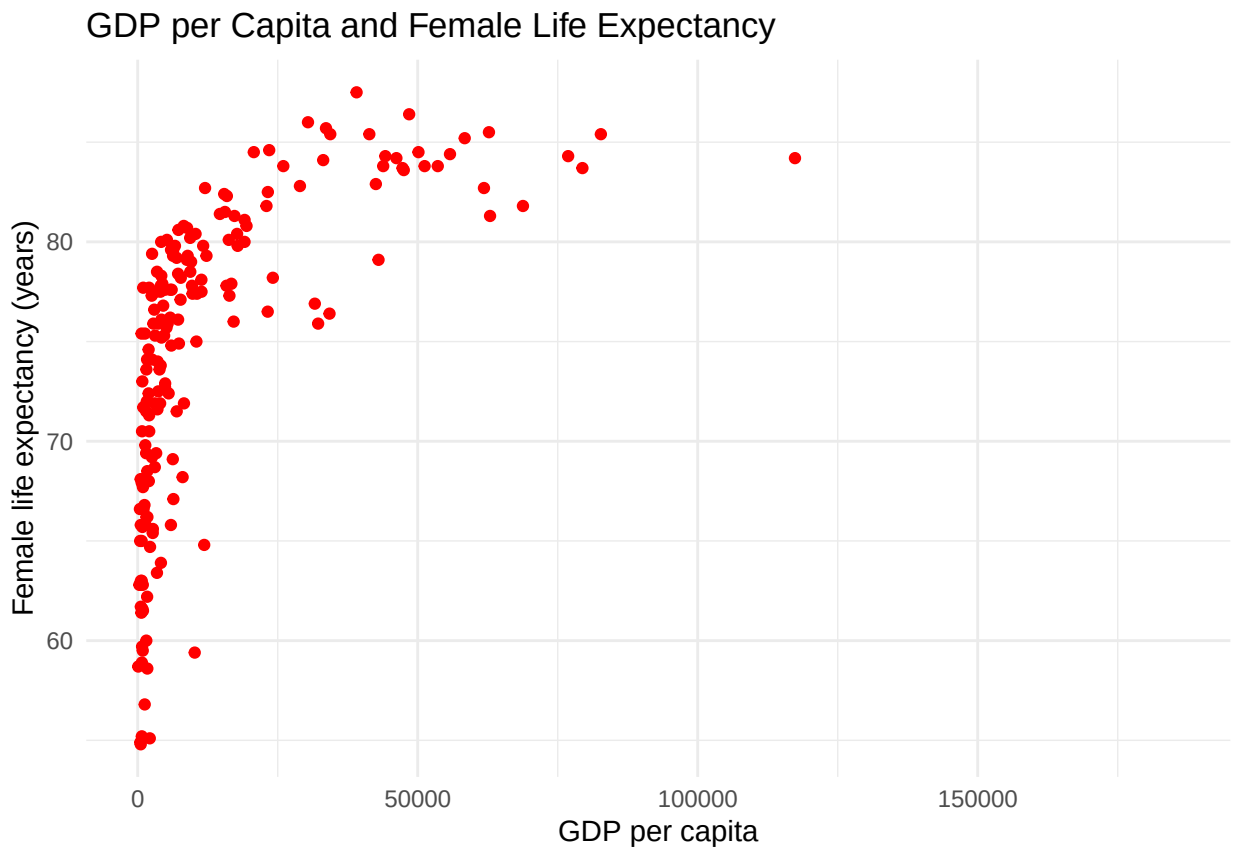
Figure 3: Distribution of infant mortality across countries

II. GDP per capita and life expectancy

Is GDP per capita associated with life expectancy?

The relationship between GDP per capita and life expectancy was examined using scatter plots and Spearman's rank correlation. GDP per capita showed a strong positive association with both female and male life expectancy, with $p = 0.842$ for females and $p = 0.830$ for males ($p < 0.001$ for both).

This indicates that countries with higher GDP per capita generally tended to have higher life expectancy for both sexes. The scatter plot also suggested that the relationship became less pronounced at higher GDP per capita levels. However, the findings indicate association rather than causation and do not establish that higher GDP per capita directly causes longer life expectancy.



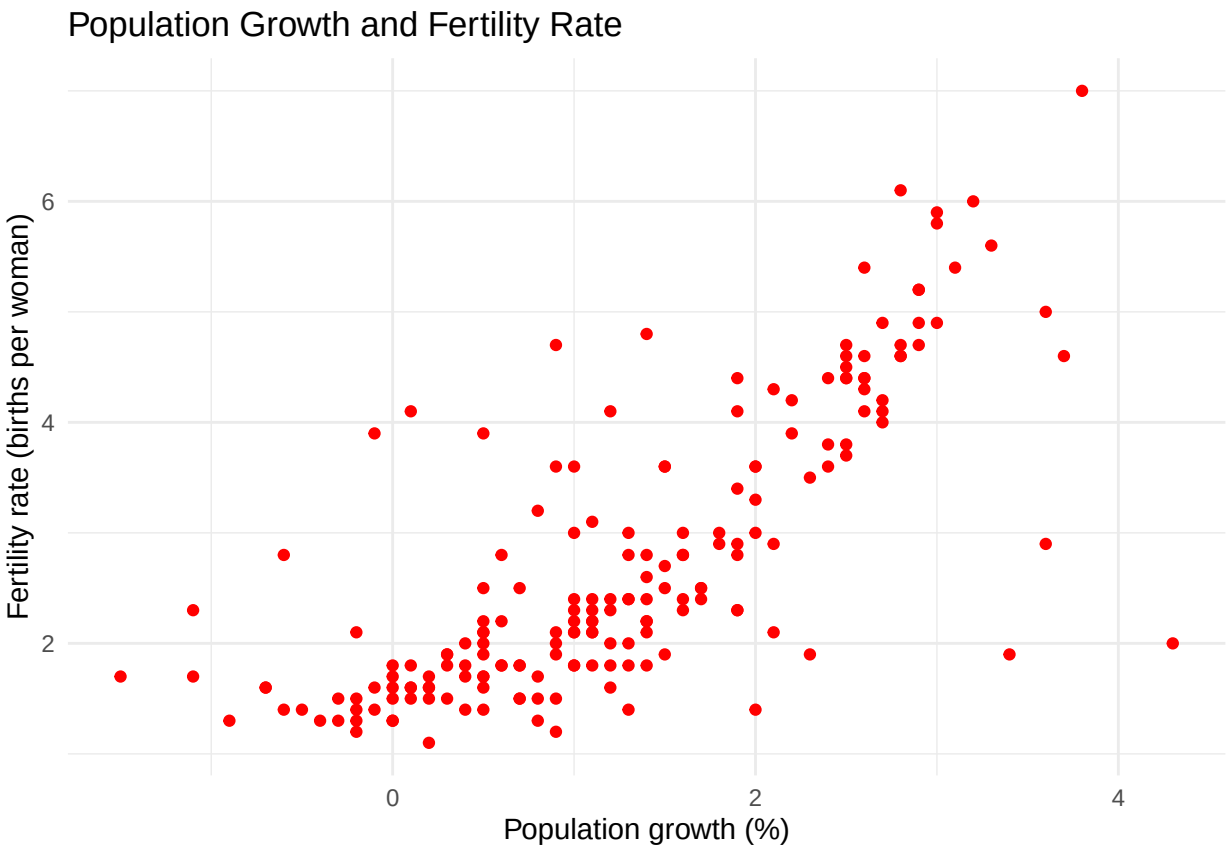

```
##
##  Spearman's rank correlation rho
##
## data:  analysis_data$gdp_per_capita and analysis_data$life_expectancy_female
## S = 172039, p-value < 2.2e-16
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##       rho
## 0.8421427
```

```
##
##  Spearman's rank correlation rho
##
## data:  analysis_data$gdp_per_capita and analysis_data$life_expectancy_male
## S = 185517, p-value < 2.2e-16
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##       rho
## 0.8297757
```

III. Is population growth associated with fertility rate?

The relationship between population growth and fertility rate was examined using a scatter plot and Spearman's rank correlation. The results showed a strong positive association, with fertility generally increasing as population growth increased. Spearman's correlation was $p = 0.780$ ($p < 0.001$), indicating strong statistical

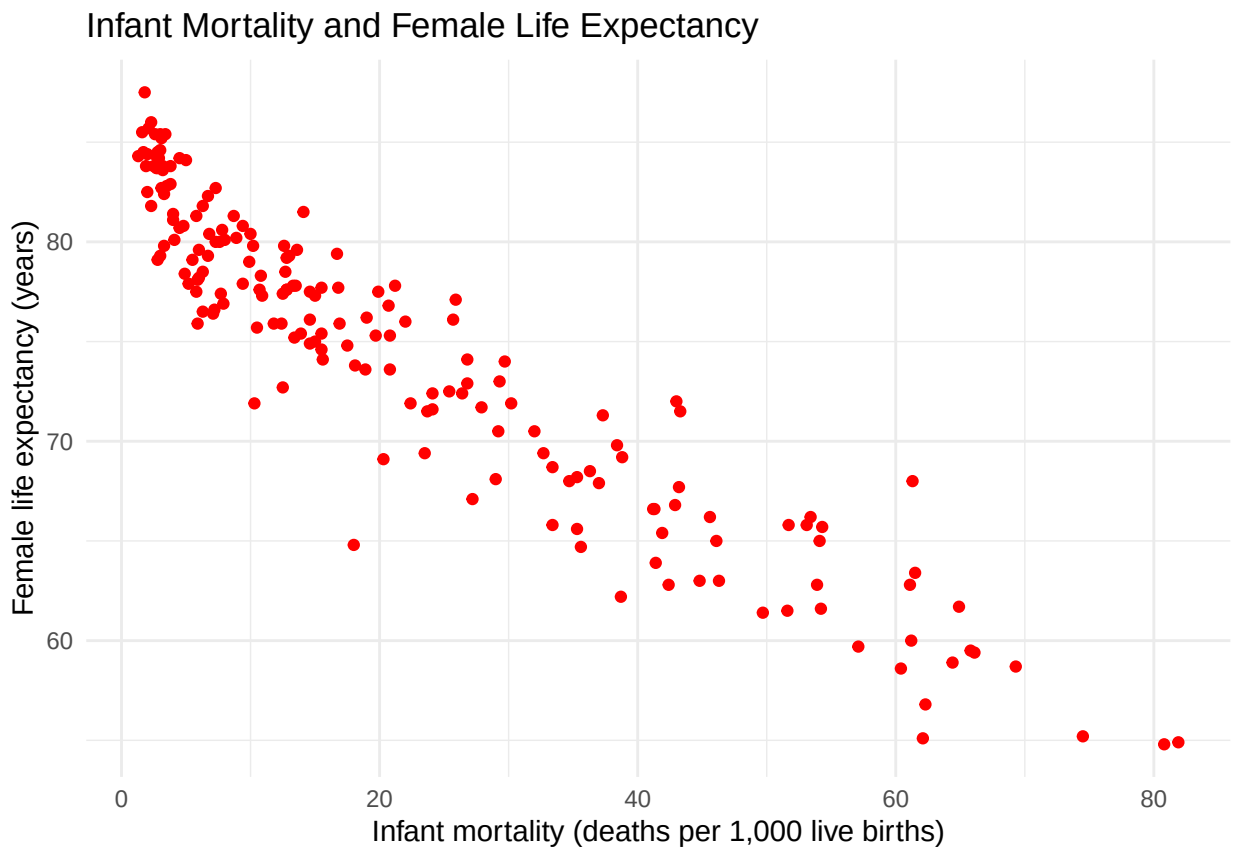
evidence of an association between the two variables. However, this correlation does not establish causation, as other demographic and socioeconomic factors may influence both population growth and fertility rates.




```
##
##  Spearman's rank correlation rho
##
## data:  analysis_data$pop_growth and analysis_data$fertility
## S = 239654, p-value < 2.2e-16
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##       rho
## 0.7801009
```

IV. Relationship between infant mortality and life expectancy

The relationship between infant mortality and life expectancy was examined using a scatter plot and Spearman's rank correlation. The results showed a very strong negative monotonic association: countries with higher infant mortality tended to have lower life expectancy. Spearman's correlation was P = −0.940 for female life expectancy and p = −0.899 for male life expectancy (n = 185 countries for each; p < 0.001 for both). The scatter plot suggested that life expectancy declined most steeply at low infant mortality levels and more gradually at higher levels. As with the other analyses, this indicates association rather than causation, since infant mortality and life expectancy are both influenced by factors such as healthcare access and income.




```
##
##  Spearman's rank correlation rho
##
## data:  analysis_data$infant_mortality and analysis_data$life_expectancy_female
## S = 2047044, p-value < 2.2e-16
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##       rho
## -0.9398847
```

```
##
##  Spearman's rank correlation rho
##
## data:  analysis_data$infant_mortality and analysis_data$life_expectancy_male
## S = 2003994, p-value < 2.2e-16
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##       rho
## -0.8990884
```

```
## [1] 185
```

```
## [1] 185
```

V. How does life expectancy differ between male and female populations?

Male and female life expectancy were compared using a paired analysis, with the female minus male difference calculated for each country. The Shapiro-Wilk test was used to assess the normality of the differences. As the differences were not normally distributed, the Wilcoxon signed-rank test was used, together with the Hodges-Lehmann estimate and 95% confidence interval of the median difference.

```
##             n female_higher         equal   male_higher      min_diff
##         187.0         187.0           0.0           0.0           0.6
##   median_diff            Q1            Q3      max_diff
##           4.4           3.3           6.1          11.8
```

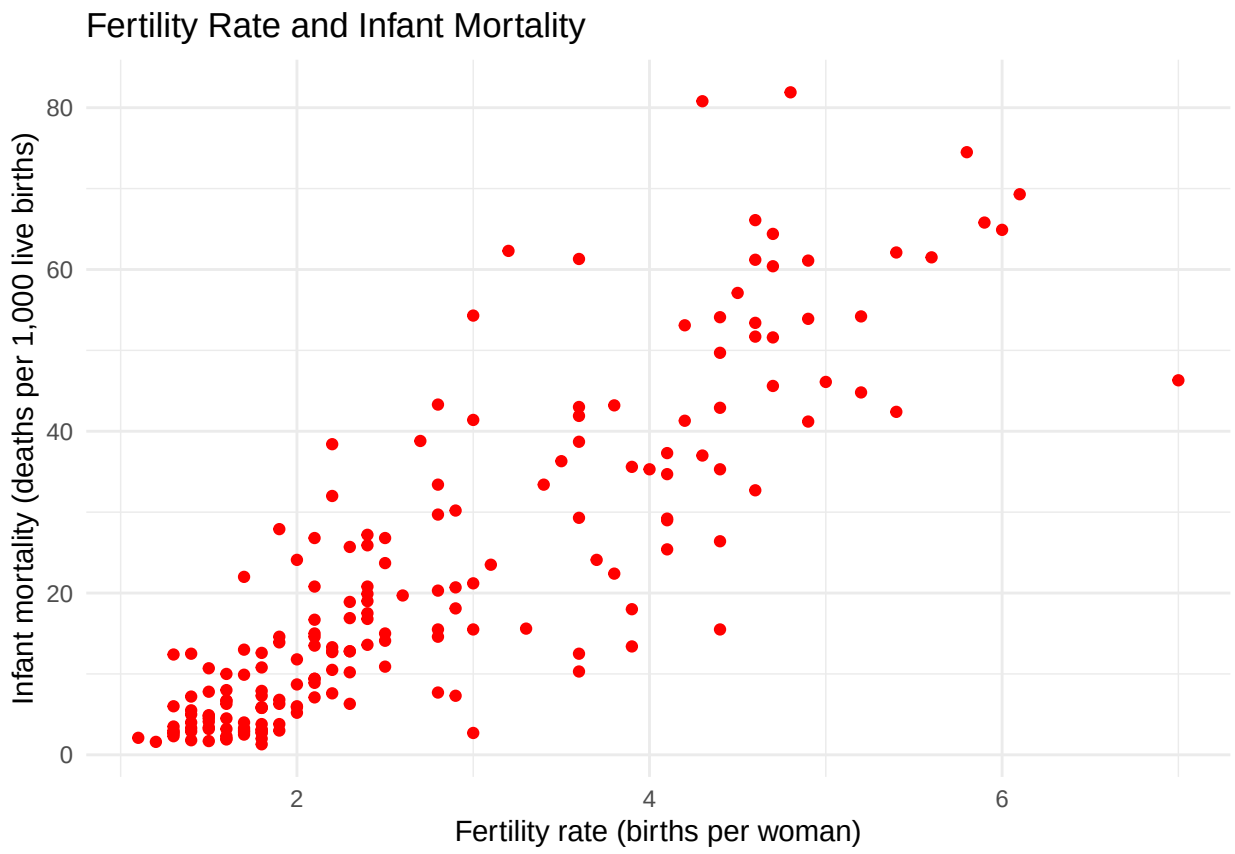
```
##
##  Shapiro-Wilk normality test
##
## data:  d
## W = 0.96196, p-value = 5.979e-05
```

```
##
##  Wilcoxon signed rank test with continuity correction
##
## data:  life_expectancy_comparison$life_expectancy_female and life_expectancy_comparison$life_expectar
## V = 17578, p-value < 2.2e-16
## alternative hypothesis: true location shift is not equal to 0
## 95 percent confidence interval:
##  4.350026 5.000075
## sample estimates:
## (pseudo)median
##       4.650026
```

VI. Fertility rate and infant mortality

Is fertility rate associated with infant mortality?

The relationship between fertility rate and infant mortality was examined using a scatter plot and Spearman's rank correlation. The results showed a strong positive association, with infant mortality generally increasing as fertility rate increased. Spearman's correlation was $p = 0.867$ ($p < 0.001$), indicating strong statistical evidence of an association between the two variables. However, this result does not establish causation, as other demographic, socioeconomic and health-related factors may influence both fertility and infant mortality.




```
##
##  Spearman's rank correlation rho
##
## data:  analysis_data$fertility and analysis_data$infant_mortality
## S = 140680, p-value < 2.2e-16
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##       rho
## 0.8666843
```

Limitations

Several limitations were considered when interpreting the findings. First, the analysis uses country-level observations, meaning that relationships observed between countries cannot necessarily be interpreted as relationships between individuals. Second, the analysis is primarily correlational and does not control for confounding variables. Third, some variables contain missing observations, so the number of countries contributing to individual analyses may differ. Finally, the dataset provides a cross-sectional view of the countries, limiting the ability to examine changes over time or establish causal relationships.

Conclusion

This analysis examined six statistical questions concerning life expectancy, GDP per capita, fertility, infant mortality and population growth across countries. The results showed substantial variation in health and demographic indicators. GDP per capita was strongly positively associated with male and female

life expectancy, while fertility was strongly positively associated with infant mortality and population growth.Female life expectancy exceeded male life expectancy in every country with data for both sexes, with a median within-country difference of 4.4 years.

Overall, the findings demonstrate important associations between economic, demographic and health indicators across countries. However, because the analysis uses country-level observational data, the associations do not establish causation. Missing observations also mean that the number of countries included may differ between analyses.

Appendix: R Code

```r
library(tidyverse)

country_data <- read_csv("country_data.csv")
```

```
## Rows: 204 Columns: 38
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr  (5): currency, iso2, capital, name, region
## dbl (33): gdp, sex_ratio, surface_area, life_expectancy_male, unemployment, ...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
country_data_clean <- country_data %>%
  distinct()

analysis_data <- country_data_clean %>%
  select(
    name,
    region,
    gdp_per_capita,
    life_expectancy_male,
    life_expectancy_female,
    fertility,
    infant_mortality,
    pop_growth
  )
```

```r
# Check the number of duplicate observations
sum(duplicated(country_data))

# Check the number of observations after removing duplicates
nrow(country_data_clean)

# Check missing values in the selected variables
colSums(is.na(analysis_data))

# Check for impossible or unusual values
sum(analysis_data$fertility < 0, na.rm = TRUE)
sum(analysis_data$infant_mortality < 0, na.rm = TRUE)
sum(analysis_data$gdp_per_capita < 0, na.rm = TRUE)
```

```r
sum(abs(analysis_data$pop_growth) > 20, na.rm = TRUE)
sum(analysis_data$life_expectancy_male < 0, na.rm = TRUE)
sum(analysis_data$life_expectancy_female < 0, na.rm = TRUE)

# key variables

variables <- data.frame(
  Variable = c(
    "name",
    "region",
    "pop_growth",
    "life_expectancy_male",
    "life_expectancy_female",
    "infant_mortality",
    "fertility",
    "gdp_per_capita"
  ),
  `What it represents` = c(
    "Country",
    "Geographical region",
    "Population growth rate",
    "Male life expectancy",
    "Female life expectancy",
    "Infant mortality rate",
    "Fertility rate",
    "GDP per person"
  ),
  `Role / Type` = c(
    "Identifier",
    "Categorical",
    "Numerical",
    "Numerical",
    "Numerical",
    "Numerical",
    "Numerical",
    "Numerical"
  )
)

kable(
  variables,
  caption = "Description of Variables"
)
```

```r
# Objective 1: Distributions of health and demographic indicators

desc_table <- analysis_data %>%
  select(
    `Male life expectancy (years)`   = life_expectancy_male,
    `Female life expectancy (years)` = life_expectancy_female,
    `Fertility (births per woman)`   = fertility,
    `Infant mortality (per 1,000)`   = infant_mortality,
    `Population growth (%)`          = pop_growth,
```

```
    `GDP per capita`                = gdp_per_capita
  ) %>%
  pivot_longer(everything(), names_to = "Variable", values_to = "value") %>%
  mutate(Variable = factor(Variable, levels = unique(Variable))) %>%
  group_by(Variable) %>%
  summarise(
    n      = sum(!is.na(value)),
    Mean   = mean(value, na.rm = TRUE),
    SD     = sd(value, na.rm = TRUE),
    Median = median(value, na.rm = TRUE),
    IQR    = IQR(value, na.rm = TRUE),
    Min    = min(value, na.rm = TRUE),
    Max    = max(value, na.rm = TRUE)
  )

knitr::kable(desc_table, digits = 1,
             caption = "Summary statistics of the analysed variables")
```

```
life_expectancy_long <- analysis_data %>%
  select(life_expectancy_male, life_expectancy_female) %>%
  pivot_longer(everything(), names_to = "Sex", values_to = "Life_Expectancy") %>%
  mutate(Sex = recode(Sex, life_expectancy_male = "Male",
                      life_expectancy_female = "Female"))

ggplot(life_expectancy_long, aes(x = Life_Expectancy, fill = Sex)) +
  geom_density(alpha = 0.4, na.rm = TRUE) +
  labs(x = "Life expectancy (years)", y = "Density", fill = "Sex") +
  theme_minimal()

# fertilty rate

ggplot(analysis_data, aes(x = fertility)) +
  geom_histogram(bins = 20, na.rm = TRUE, fill = "steelblue", colour = "white") +
  labs(x = "Fertility rate (births per woman)", y = "Number of countries") +
  theme_minimal()

# infant mortality

ggplot(analysis_data, aes(x = infant_mortality)) +
  geom_histogram(bins = 20, na.rm = TRUE, fill = "steelblue", colour = "white") +
  labs(x = "Infant mortality (deaths per 1,000 live births)", y = "Number of countries") +
  theme_minimal()
```

```
# Objective 2: GDP per capita and life expectancy

ggplot(
  analysis_data,
  aes(x = gdp_per_capita, y = life_expectancy_female)
) +
  geom_point(na.rm = TRUE) +
  labs(
    title = "GDP per Capita and Female Life Expectancy",
    x = "GDP per capita",
```

```
    y = "Female life expectancy (years)"
  ) +
  theme_minimal()

cor.test(
  analysis_data$gdp_per_capita,
  analysis_data$life_expectancy_female,
  method = "spearman",
  exact = FALSE
)

cor.test(
  analysis_data$gdp_per_capita,
  analysis_data$life_expectancy_male,
  method = "spearman",
  exact = FALSE
)


# Objective 3: Population Growth and Fertility Rate

ggplot(
  analysis_data,
  aes(x = pop_growth, y = fertility)
) +
  geom_point(na.rm = TRUE, color = "RED") +
  labs(
    title = "Population Growth and Fertility Rate",
    x = "Population growth (%)",
    y = "Fertility rate (births per woman)"
  ) +
  theme_minimal()

cor_test_growth <- cor.test(
  analysis_data$pop_growth,
  analysis_data$fertility,
  method = "spearman",
  exact = FALSE
)

cor_test_growth


# Objective 4: Relationship between infant mortality and life expectancy

ggplot(analysis_data, aes(x = infant_mortality, y = life_expectancy_female)) +
  geom_point(na.rm = TRUE, color = "RED") +
  labs(
    title = "Infant Mortality and Female Life Expectancy",
    x = "Infant mortality (deaths per 1,000 live births)",
    y = "Female life expectancy (years)"
  ) +
  theme_minimal()
```

```r
cor.test(analysis_data$infant_mortality, analysis_data$life_expectancy_female,
         method = "spearman", exact = FALSE)

cor.test(analysis_data$infant_mortality, analysis_data$life_expectancy_male,
         method = "spearman", exact = FALSE)

# n used in each test
sum(complete.cases(analysis_data$infant_mortality, analysis_data$life_expectancy_female))
sum(complete.cases(analysis_data$infant_mortality, analysis_data$life_expectancy_male))
```

```r
# Objective 5: Male and female life expectancy

life_expectancy_comparison <- analysis_data %>%
  mutate(difference = life_expectancy_female - life_expectancy_male)

d <- na.omit(life_expectancy_comparison$difference)

# Counts and summary of the within-country differences
c(n = length(d),
  female_higher = sum(d > 0),
  equal = sum(d == 0),
  male_higher = sum(d < 0),
  min_diff = min(d),
  median_diff = median(d),
  Q1 = unname(quantile(d, 0.25)),
  Q3 = unname(quantile(d, 0.75)),
  max_diff = max(d))

# Normality of the differences
shapiro.test(d)

# Wilcoxon signed-rank test with Hodges-Lehmann estimate and 95% CI
wilcox.test(life_expectancy_comparison$life_expectancy_female,
            life_expectancy_comparison$life_expectancy_male,
            paired = TRUE, exact = FALSE, conf.int = TRUE)

# Objective 6: Fertility rate and infant mortality

ggplot(
  analysis_data,
  aes(x = fertility, y = infant_mortality)
) +
  geom_point(na.rm = TRUE, color = "RED") +
  labs(
    title = "Fertility Rate and Infant Mortality",
    x = "Fertility rate (births per woman)",
    y = "Infant mortality (deaths per 1,000 live births)"
  ) +
  theme_minimal()

cor_test_fertility <- cor.test(
  analysis_data$fertility,
  analysis_data$infant_mortality,
```

```
  method = "spearman",
  exact = FALSE
)

cor_test_fertility
```